MIL-H-1987G
27 June 1980
SUPERSEDING
MIL-H-1987F
23 June 1977

MILITARY SPECIFICATION

HELMET, FIRE FIGHTER'S

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a fire fighter's helmet.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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DRAWING

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

8-1-25 - Helmet, Fire Fighter's

(Miniature copy of Drawing 8-1-25 shown as figure 2, is attached for information purposes only).

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN NATIONAL STANDARDS INSTITUTE, INC.

Z87.1-1968 - Practice for Occupational and Educational Eye and Face Protection
Z89.1-1969 - Safety Requirements for Industrial Head Protection

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 1972-1979 - Structural Fire Fighters Helmets

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

3. REQUIREMENTS

* 3.1 First article. The contractor shall furnish samples for first article inspection and approval (see 4.3 and 6.3).

* 3.2 Materials and components. Except where otherwise specified herein all materials (see 6.5) and components used in the construction of the helmet shall be at the option of the contractor. All materials coming in contact with the wearers head shall be non-irritating to normal skin (see 4.4.1.1).

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3.2.1 Metal components. All metal components shall be corrosion-resistant or corrosion-resistant coated and free of sharp edges or burrs.

3.2.2 Face shield. Unless otherwise specified (see 6.2) the helmet shall be furnished with a clear transparent polycarbonate plastic face shield. The face shield shall be attached to the helmet in such a manner that it can be pivoted and secured in either an up or down position. The face shield shall meet the optical requirements specified in paragraphs 6.3.2.1 and 6.3.2.2 of ANSI Z87.1-1968 (see 4.4.1.1). The configuration of the face shield shall be the manufacturer's commercial design.

3.2.3 Ear and neck flap. Unless otherwise specified (see 6.2) the helmet shall be furnished with a removable ear and neck flap of the manufacturer's commercial design. The exterior color shall approximate the color of the helmet shell. Other colors may be used with the approval of the contracting officer.

* 3.2.4 Headband and crown straps.

* 3.2.4.1 Headband. The headband shall be adjustable in 1/8 inch (3 mm) increments. The minimum size range shall be 6-1/2 to 8 (standard commercial designation) and shall be obtainable by the use of one or more headbands. The size range and adjustment points shall be marked on the headband in a permanently legible manner. When the headband is adjusted to the maximum size the headband shall not contact the shell except at attachment points. The surface of the headband, in contact with the wearer's head, shall not be less than one-inch (25 mm) wide. The headband(s) shall be removable and replaceable.

* 3.2.4.2 Crown straps. When crown straps are provided in the contractors helmet design, the straps shall be adjustable to provide proper fit and maintain proper head height. When properly adjusted the straps shall be in contact with the wearers head during normal use.

* 3.2.5 Sweatband. The sweatband may be of the removeable-replaceable type or may be integral with the headband. The sweatband shall cover as a minimum the forehead portion of the headband. When the headband is adjusted to the maximum size the sweatband shall not contact the shell when the helmet is properly postioned on the wearers head.

3.2.6 Color.

3.2.6.1 Helmet shell. The color of the helmet shell shall be lime yellow. The shell shall be pigmented with the required color during the molding process if feasible. Surface coating finishes may be applied where integral color is not feasible.

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3.2.6.2 Chin strap. The color of the chin strap shall be neutral or a dark color.

3.2.6.3 Sweatband. The color of the sweatband shall be neutral or a dark color.

* 3.3 Construction. The construction of the helmet shall conform to the requirements of NFPA 1972-1979, Drawing 8-1-25 and as specified herein.

3.3.1 Weight. The weight of the helmet complete with energy absorbing material and all components except the face shield and ear and neck flap shall not exceed 35 ounces (992 g).

3.3.2 Finish.

3.3.2.1 Helmet shell. The outer and inner surfaces of the helmet shell shall have a uniform high luster. The brim edge shall be smooth with an even radius and free from burrs and sharp edges and may have an edge trim. The helmet shall be free of chips, cracks, blisters or voids.

3.3.2.2 Metal components. Metal components with painted surfaces shall be free from flaking, peeling, orange peel, sags, runs or scratches exposing base metal.

3.3.3 Setting of snap fasteners, eyelets, and rivets. All snap fasteners, eyelets, and rivets used in the construction of the helmet shall be securely clinched or peened, as applicable, in such a manner that the components will be securely joined and without damaging the adjacent materials.

3.3.4 Stitching. When stitching is used in the construction of the helmet suspension system the following shall apply:

a. Ends of all stitching shall be backstitched or overstitched 1/2 inch (13 mm) minimum.

b. Thread breaks, or two or more consecutive skipped or runoff stitches shall be repaired by overstitching 1/2 inch (13 mm) minimum in each direction beyond the defective stitching area.

c. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

d. All thread ends shall be trimmed to a length of not more than 1/4 inch (6 mm).

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3.3.5 Fusing of ends of nylon webbing. If nylon webbing is used in the construction of the helmet suspension system all exposed ends that are not turned under shall be fused to prevent fraying. The apparatus used to fuse the webbing ends shall be capable of providing sufficient heat to fuse the cut ends of the webbing yarns together.

3.3.6 Asssembly of components. All components of the helmet, i.e., suspension assembly, sweatband, chinstrap, face shield, ear and neck flap, etc. shall be installed in the helmet.

* 3.4 Performance. The helmet shall meet the performance requirements specified herein and NFPA 1972-1979 except as follows:

a. Delete paragraphs 3-1 and 3-1.1, Top Impact Requirement-Force Transmission.

b. Paragraph 3-2.1, at the end of the paragraph add "The helmet shall show no evidence of cracking or fracture as a result of impact testing."

c. Table 3-2.1, maximum acceleration for the top impact location shall be 100 x Gn, 981 m/sec/sec; 3220 ft/sec/sec in lieu of the specified requirements.

3.4.1 Suspension assembly retention. The average minimum force required to separate the suspension assembly from the helmet shell shall be not less than 15 pounds (6.8 kg) when tested as specified in 4.4.2.2. The minimum force required to separate any individual attachment point of the suspension assembly from the helmet shell shall be 10 pounds (4.5 kg).

3.4.2 Fasteners and adjusting mechanisms durability. All fasteners and adjusting mechanisms used in the helmet suspension system shall perform satisfactorily without any visible deterioration when tested as specified in 4.4.2.2.

3.4.3 Water absorption. The helmet shell assembly consisting of the helmet shell and energy absorbing lining shall meet the water absorption requirement specified for helmets conforming to ANSI Z89.1-1969.

3.4.4 Ear and neck flap flame resistant. The ear and neck flap shall meet the flame resistance requirement specified in NFPA 1972-1979 for ear covers.

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* 3.5 Identification markings and instructions. The identification markings listed below shall be permanently stamped on each headband assembly in characters 1/8 inch (3 mm) high minimum. The markings shall be clearly legible and shall not show smearing, bleeding or offsetting. As an option, all or part of the identification marking requirements specified in paragraph 2-2.1 of NFPA 1972-1979 may also be marked on the helmet. The instructions and the warning marking required in paragraphs 2-2.2 and 2-2.4 of NFPA 1972-1979 shall apply. The instructions required in 2-2.2 shall include instructions as to the proper procedure for adjusting and securing the helmet to the users head.

Nomenclature
Contract No.
Stock No.
Contractor

3.6 Replacement of defective components. During the cutting and manufacturing process, textile components having material defects or damages that are classified as defects in 4.4.2.1.1 shall be removed from production and replaced with non-defective components.

3.7 Workmanship. The finished helmet shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. The first article submitted in accordance with 3.1 shall be examined for the defects specified in 4.4.2.1.1, 4.4.2.1.2 and 4.4.2.1.3 and tested for the characteristics specified in table I. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

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* 4.4 Quality conformance inspection. Sampling for inspection shall be in accordance with MIL-STD-105, except where otherwise indicated hereinafter. Paragraphs 4-3 through 4-3.3, 4-4 and 4-4.1 of NFPA 1972-1979 shall not apply.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.1.1 Certification. The contractor shall furnish a certificate of compliance certifying that all materials used in the helmet that will come in contact with the user's head are non-irritating to normal skin and that the face shield conforms to the specified optical requirements.

* 4.4.2 End item inspection.

4.4.2.1 End item examination. The end item shall be examined for visual dimensional and weight defects in accordance with 4.4.2.1.1, 4.4.2.1.2 and 4.4.2.1.3. The applicable inspection levels and acceptable quality levels shall be as indicated in 4.4.2.1.4. The sample unit shall be one completely fabricated helmet and the lot size shall be expressed in units of single helmets.

* 4.4.2.1.1 Visual examination. The classification of defects shall be in accordance with the list below.

Examine	Defect	Classification Major	Minor
Suspension system	Any cut, tear or hole	X	
	Width of chin strap or headband less than specified		X
	Headband or sweatband not replaceable type, i.e., cannot be removed	X	
	Headband cannot be adjusted within the specified range	X	
	Exposed ends of nylon webbing not fused (when applicable)		X
	Color of any component not as specified		X
Ear and neck flap	Not removable	X	
	Not color specified		X

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Examine	Defect	Classification Major	Minor
Hardware	Broken or malformed, corroded, sharp edge or burr	X	
	Any component improperly installed causing failure to serve intended purpose	X	
	Painted components: any flaking, peeling, orange peel, sags, runs or scratches exposing base metal		X
	Any snap fastener, eyelet, or rivet insecurely clinched or peened to the degree that separation may occur from the assembly	X	
	Any snap fastener, eyelet or rivet excessively clinched or peened to the degree that materials are damaged	X	
	Any snap fastener that does not function properly, i.e., fails to: snap closed, provide a secure closure or open freely	X	
Shell	Color not as specified or not uniform		X
	Any sharp edge or burr		X
	Any hole (other than required for assembly of suspension system to shell), crack, chip, soft spot, blister or void	X	
Face shield	Not adjustable to an up and down position or can not be secured in set position	X	
	Any pit, bubble, scratch, void or blister	X	
	Any crack, hole or break	X	
	Not clear transparent	X	

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Examine	Defect	Classification Major	Minor
Stitching (if used in construction of helmet suspension assembly)	Stitch tension loose resulting in loose top or bobbin thread or excessively tight resulting in puckering of materials		X
	Ends of stitching not secured as specified		X
	Thread breaks, or two or more consecutive skipped or runoff stitches not overstitched		X
	Thread ends not trimmed as specified		X
Components and assembly (applicable unless otherwise classified herein)	Any component omitted, malformed, or misplaced	X	
	Any component not firmly affixed or attached to an extent that any component may become separated during use	X	
	Design of any component not as specified	X	
	Any unauthorized repair	X	
	Any component not installed in the helmet, e.g., chin strap not attached	X	
Identification and warning markings	Omitted, illegible, incorrect, or size of characters not as specified		X
Instructions sheet	Omitted, illegible or incomplete		X
Cleanness	Grease or oil stains		X
	Thread ends not trimmed		X

* 4.4.2.1.2 Dimensional examination. An examination shall be made to determine that the overall dimensions of the helmet conform to the dimensional requirements specified. Any dimension exceeding the specified requirement shall be classified as a defect.

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4.4.2.1.3 Weight examination of the helmet assembly. The helmet assembly without face shield and ear and neck flap attached shall be examined for the specified weight. The helmet shall be weighed on a balance or scale graduated to within not more than 1/4 ounce (10 g). Any helmet exceeding the weight limitation shall be classified as a defect.

4.4.2.1.4 Inspection levels and acceptable quality levels. The inspection levels and the acceptable quality levels (AQLs) expressed in defects per one hundred units shall be as follows:

Examination paragraph	Inspection level	AQL	
		Major	Total
4.4.2.1.1	II	2.5	6.5
4.4.2.1.2	S-3	---	4.0 (one class)
4.4.2.1.3	S-3		1.5 (one class)

4.4.2.2 End item testing. The complete helmet assembly shall be tested for the characteristics listed in table I. The sample unit shall be as indicated in table I. For the suspension assembly retention test and the test of the fasteners and adjusting mechanisms for durability the inspection level shall be S-2 and the acceptable quality level shall be 2.5 defects per one hundred units. For all other testing the sample size for the designated lot size shall be as indicated below. Any test failure shall be cause for rejection of the lot.

Lot size	Sample size
1000 or less	1
1001 up to 2000 1/	2

1/ For lot sizes greater than 2000 the sample size shall be increased by one for each additional quantity of 1000 items or fraction thereof.

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TABLE I. End item tests

Characteristic	Specification reference: Requirement	Specification reference: Test method	Requirements applicable to: Sample unit	Number of determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Suspension assembly retention	3.4.1	4.5.1	X	1/		0.25 pound (0.1 kg)	1 helmet
Retention system	3.4	NFPA 1972-1979	X	1	X		1 helmet
Durability of fasteners and adjusting mechanisms	3.4.2	4.5.2	X	2/	X		1 helmet
Water absorption	3.4.3	ANSI-Z89.1-1969	X	1		0.1 percent	3 helmets
Flame resistance of helmet shell	3.4	NFPA 1972-1979	X	1	X		1 helmet
Impact-acceleration/ Impact resistance	3/	4/ 6/	X	20	X		4 helmets
Penetration resistance	3.4	NFPA 1972-1979 5/ 6/	X	8	X		4 helmets
Heat resistance	3.4	NFPA 1972-1979	X	1	X		1 helmet

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TABLE I. End item tests (cont'd)

Characteristic	Specification reference: Requirement	Specification reference: Test method	Requirements applicable to: Sample unit	Number of determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Electrical insulation	3.4	NFPA 1972-1979	X	1	X		1 helmet
Flame resistance of ear and neck flap	3.4.4	NFPA 1972-1979	X	1	X		1 helmet

1/ For each sample unit a determination shall be made for each attachment point of the suspension assembly to the helmet shell and a determination of the average of all the individual determinations.

2/ For each sample unit a determination shall be made for each fastener or adjusting mechanism of the helmet suspension assembly.

3/ The requirement for impact-acceleration/impact resistance shall be as specified in NFPA 1972-1979, except as changed in 3.4.b and 3.4.c.

4/ Testing for impact-acceleration shall be in accordance with NFPA 1972-1979 except that the top impact site shall be within a 2-inch (51 mm) radius from the intersection of the midsaggital plane and the bitragion-coronal arc extended to the surface of the helmet. The front, rear and side impact centers shall be 1 +1/2, -0 inches (25 +13, -0 mm) above the reference plane. The test anvil shall be steel. The striking face shall be a spherical segment with a radius of 1.9 $\pm$ 0.3 inches (48 $\pm$ 8 mm). The test shall be considered a failure if one or more impact-acceleration tests exceeds the maximum acceleration specified in NFPA 1972-1979 or if cracking or fracturing of the helmet occurs. (In the event of a dispute as to whether the helmet shell is cracked or fractured the helmet shell shall be subjected to and pass the electrical insulation test.) A different helmet shall be used for impact resistance testing for each of the four exposure conditions.

5/ A different helmet shall be used for penetration resistance testing for each of the four exposure conditions.

6/ The headband shall be adjusted to obtain a snug fit with the head form. If the helmet design includes crown straps the straps shall be properly adjusted to obtain contact with the top of the head form.

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* 4.4.3 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination of closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Marking (exterior and unit pack)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged or not as specified
Workmanship	Inadequate application of components, such as: incomplete closure of container flap, loose strapping, improper taping, inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.4 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per 100 units.

Examine	Defect
Finished dimension	Length, width or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application

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4.5 Methods of inspection.

4.5.1 Suspension assembly retention. The helmet assembly shall be positioned and secured so that all attachment points of the suspension assembly are on a horizontal plane. Each attachment point of the suspension assembly to the helmet shell shall be tested by slowly applying a pull force in the direction required to release the mechanism with sufficient force to separate the attachment point from the helmet shell. Suggested techniques for applying the load are shown in figure 1. The force gauge shall be accurate to 0.25 pounds (0.1 kg). Care shall be taken to apply the pull force through the centerline at each attachment point. If the pull force is applied at any other angle the test results shall be unacceptable and the test shall be repeated. The individual forces and the average force shall be reported. Failure of the suspension assembly retention system to meet the requirements of 3.4.1 shall constitute failure of the test.

4.5.2 Fasteners and adjusting mechanisms durability. Each fastener or adjusting mechanism of the helmet suspension assembly shall be opened and closed or secured and unsecured, as applicable, ten times. Any fastener or adjusting mechanism that indicates any evidence of deterioration of effectiveness by the tenth cycle shall constitute a failure.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

* 5.1.1 Level A. One completely assembled helmet with all components attached and with instruction sheet, (see 3.5) shall be unit packed in a snug-fitting fiberboard box conforming to style RSC (end loading), grade 200 of PPP-B-636. Each box shall be closed in accordance with method II as specified in the appendix of the box specification, except that the inspection shall be in accordance with 4.4.3.

* 5.1.2 Commercial. The helmets shall be preserved in accordance with MIL-STD-1188.

* 5.2 Packing. Packing shall level A, B, or Commercial as specified (see 6.2).

* 5.2.1 Level A packing. Six helmets, unit packed as specified in 5.1.1, shall be packed in a snug fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A unit packs shall be packed on end two in length, three in width, and one in depth within a shipping container. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.3.

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* 5.2.2 Level B packing. Six helmets, unit packed as specified in 5.1.1, shall be packed in a snug fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A unit packs shall be packed on end, two in length, three in width, and one in depth within a shipping container. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification, except that the inspection shall be in accordance with 4.4.3.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.3.

* 5.2.3 Commercial packing. Helmets, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Palletization. When specified (see 6.2), helmets, packed as specified in 5.2, shall be palletized in accordance with type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If a container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.4 Marking. In addition to any special marking required by the contract, unit packs, shipping containers, and (when applicable) palletized loads shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

6. NOTES

6.1 Intended use. The helmets are intended for use by fire fighters.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Quantity of first article sample required (see 3.1, 4.3 and 6.3).
(c) When face shield or ear and neck flap is not required (see 3.2.2 and 3.2.3).
(d) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(e) When weather resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(f) When palletization is required (see 5.3).

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* 6.3 First article. First article samples shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for inspection and approval of the first article samples.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.5 Recycled materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

* 6.6 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

Custodians:

Army - GL
Air Force - 99
Navy - NU

Review activities:

Army - MD
Navy - YD
DLA - CT

User activity:

Air Force - 45

Preparing activity:

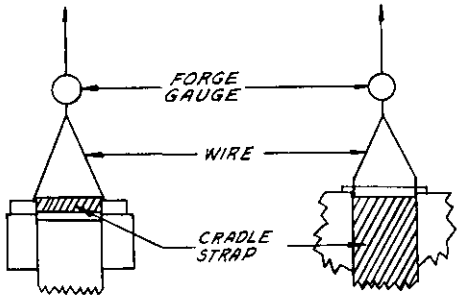
Army - GL

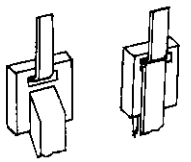
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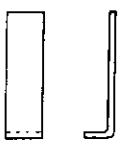
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REVISIONS				
ZONE	LTR	DESCRIPTION	DATE	APPROVED

NOTES:







HOOK:- RIGID MATERIAL APPROX 1/16" THK
DESIGN OPTIONAL TO FACILITATE
ATTACHMENT TO HELMET AND
ATTACHMENT OF FORCE GAUGE

FIGURE 1 SCHEMATICS: TECHNIQUES FOR APPLYING LOAD FOR PERFORMANCE OF SUSPENSION RETENTION TEST (TYPICAL ALL SUSPENSION ATTACHMENT POINTS)

ITEM NO.	QTY REQD	CODE IDENT	PART NO. OR IDENTIFYING NO.	NOMENCLATURE OR DESCRIPTION	MATERIAL	MATL SPEC	DWG NO.

PARTS LIST

NEXT ASSY	USED ON
APPLICATION	

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
WIRE AND METAL GAGES EXCLUDED

TOLERANCES:
FRACTIONS ±
ANGLES ±
2 PLACE DIMS. ±
3 PLACE DIMS. ±

MATERIAL

DATE 19 JUN 75
PREPARED R.P. WHITH
CHECKED
ENGINEER

APPROVAL DESIGN ACTIVITY
Richard Q. Elwell
APPROVAL OTHER

U.S. ARMY NATICK LABORATORIES
NATICK, MASS. 01760

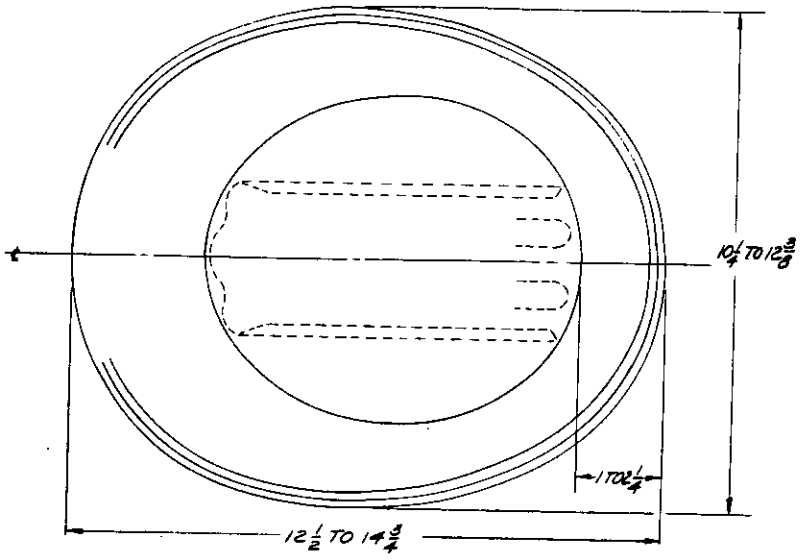
SCHEMATIC FOR HELMET SUSPENSION RETENTION TEST

SIZE	CODE IDENT NO	DRAWING NO.
C	81337	8-1-46

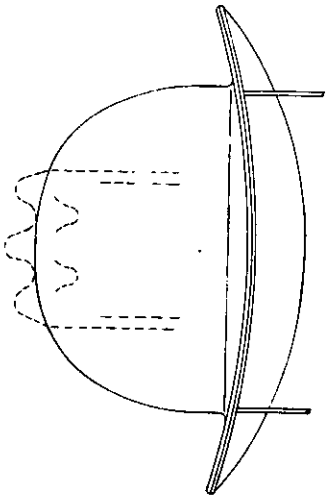
SCALE	UNIT WT.	SHEET	OF

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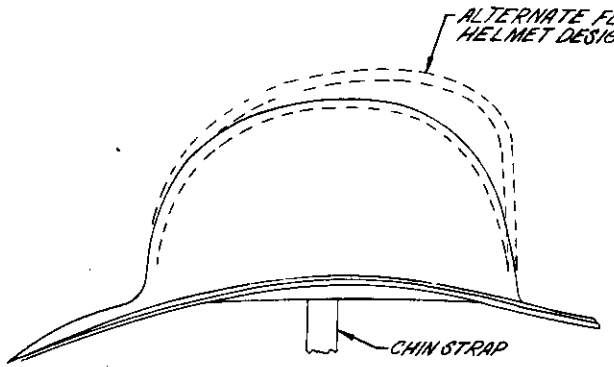
REVISIONS						
SYM	D W G NO	DR	CK	DESCRIPTION	PROJ ENGR	DATE APPROVED
A	8-306	L	FFA	REVISED AND REDRAWN	JH	15 AUG 63
B	TDO-12	R.W.	LB BAP	ADDED "CENTER ON SHELL WITHIN ¼ INCH TO "D" RING W/CLIP, DELETED NOTE & HOLES FOR SUSPENSION BAND ON FRONT VIEW & SIDE VIEW; ± 3/8 WAS ± 1/2	[illegible]	19 JUN 75
C	TDO -656	NB		DELETED "D" RING & CLIP, REVISED DIM. & NOTE		



TOP VIEW



FRONT VIEW



SIDE VIEW

NOTE:
SHELL DESIGN - ILLUSTRATIVE ONLY AND IS NOT RESTRICTIVE EXCEPT THAT ANY DIVIATIONS FROM THE GIVEN DIMENSIONS MUST BE APPROVED BY THE PROCURING AGENCY.

FIG. 2. THIS FIGURE IS A MINIATURE COPY OF THE ISSUE OF DRAWING 8-1-25 IN EFFECT ON THE DATE OF SPECIFICATION APPROVAL, AND IS FOR INFORMATION ONLY. THE ISSUE OF THE DRAWING REFERENCED IN THE CONTRACT SHALL GOVERN.

ITEM NO.	NAME	NO. REQD	MATERIAL	DWG. NO.
ITEM LIST				

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
FABRICATING TOLERANCES
(WIRE & METAL GAGES EXCLUDED)
FRACTIONS ±1/32 DECIMALS ANGLES

SUPERSEDES DRAWING:
8-1-25, 8-1-26 & 8-1-27
DATED: 3 MAY 1960

DRAWN: E. LEVERAULT
CHECKED: E.E. FILLO
SCALE: FOR REF ONLY NONE
PROJ ENGR: J. Hayden
CHIEF DIVISION: U.G. Terrill
APPROVED FOR THE Q.M.G.: [signature]
DATE: 15 AUGUST 1963
DRAFTING WORK ORDER NO: DA-306, ME-3196

DEPARTMENT OF THE ARMY
QUARTERMASTER CORPS
Q. M. RESEARCH & ENGINEERING COMMAND
NATICK, MASS.

HELMET, FIREMAN'S PLASTIC

CODE	SPEC. NO.	SIZE	DWG. NO
81937	MIL-H-19847	C	8-1-25

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FOLD

DEPARTMENT OF THE ARMY
US ARMY NATICK RESEARCH and DEVELOPMENT COMMAND
NATICK, MASSACHUSETTS 01760
OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
DRDNA-ES

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD-314



Commander
US Army Natick Research and Development Command
ATTN: DRDNA-ES
Natick, MA 01760

NATICK OP 1 AUG 78
FOLD

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NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-H-1987G; Helmet, Fire Fighter's

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

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